

Name: _____ Period: _____

Fri, May 15

Warmup

$$v = f\lambda \quad f = f_s \left(\frac{v \pm v_o}{v \mp v_s} \right) \quad \frac{1}{d_i} + \frac{1}{d_o} = \frac{1}{f} \quad M = \frac{h_i}{h_o} = \frac{-d_i}{d_o}$$

Speeds of Sound: air: 340 m/s water: 1530 m/s iron: 5100 m/s

Speed of Light: 3.0×10^8 m/s

1. A sports car is moving towards you at a speed of 22 m/s to the east. You are in front of him, driving away at 17 m/s to the east. The sports car's horn sounds a frequency of 498 Hz. What frequency do you hear?

2. What are the speeds of each of the following waves?
 - (a) the sound of a violin playing concert "A" pitch
 - (b) a red light wave
 - (c) the broadcast radio wave from WFMS.
 - (d) the screech of an owl
 - (e) the sound of thunder
 - (f) a flash of lightning

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